

The Royal Game of Ur: Virtual Reality Prototype of the Board Game Played in Ancient Mesopotamia

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ABSTRACT

We present a virtual reality implementation of the “Royal Game of Ur”, an ancient board game played by the people of Akkadian empire since around 3000 BC. The game’s rules recently deciphered from the cuneiform’s tablets have been implemented using MinMax approach with alpha-beta pruning optimization, allowing for a highly competitive game-play against even the most skilled human players. The game utilizes freehand interaction and, experimentally, a brain computer interface to roll the dice and move the pieces.

Index Terms: Human-centered computing—Interaction design; Applied computing—Media arts Computing methodologies—Virtual reality Computing methodologies—Mixed / augmented reality

1 INTRODUCTION

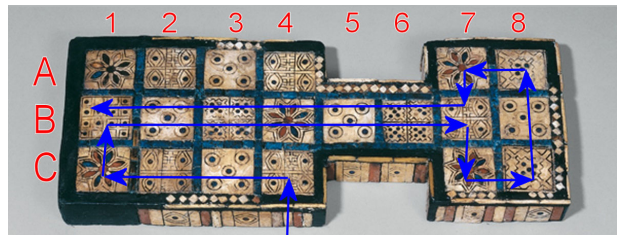


Figure 1: The path of the game. The pieces are launched from square C4 for the first player or A4 for the second player. Player moves the pieces along the path indicated by the blue arrows. The path overlap with the second player’s path when pieces are on the row B. Each piece must traverse the row B twice in order to exit the board via B1.

The Royal Game of Ur is an ancient board game that was popular among both aristocracy and common people in ancient Mesopotamia. The game itself was discovered during an excavation in the 1920s, at the Royal Cemetery of Ur on the territory of today’s Iraq [2].

It is believed that the earliest recorded rule book was written by a Babylonian astronomer who noted the game was played by “all people at that time” [3, 4]. However, the exact rules of the game remained unknown for nearly a hundred years, until they were deciphered in 2007 by Dr. Irving L. Finkel from the Babylonian cuneiform tablet [1].

Our goal in implementing the game was twofold. First and foremost, we believe that the game which survived over 5000 thousand years is worth preserving as a heritage of all humanity. Second, since Akkadians so enjoyed playing the game, we believe that modern audiences will enjoy the game, too, while learning about the ancient Mesopotamian empire.

2 RULES OF THE ROYAL GAME OF UR

In the “Royal Game of Ur”, two players are racing against each other by moving their pieces from the starting point to the finish line on the board. The game involves both an element of luck and an element of strategy. To win the game, the player needs to get all their pieces to exit the board faster than their opponent (see Fig 2).

The game set consists of the following elements:

1. The game board with twenty squares
2. Two types of tetrahedron-shaped dices for each player
3. Five different pieces for each player

Each of the tetrahedron dices play a different role. The first dice represent a choice of a number, from one to four, represented as stripes. The second dice, with two painted corners, is a boolean decision dice. Landing with colored corner on top means ‘yes’ while a landing with a blank corner on top means ‘no’.

The five pieces are color-coded to represent each player. Each piece have a face to represent five kinds of birds: the Swallow, the Storm Bird, the Raven, the Rooster, and the Eagle (Fig 2).

To start the game, both players roll the numbered dice. Whomever got the higher number goes first. The players then takes their turn to roll both dices. The number dice determines which piece can enter the game. For example, to launch the Swallow piece to the board, the player needs to roll a “2” on their numbered dice. To launch any other piece onto the board, the player must first run the Swallow piece, and then roll a number corresponding to an unused piece, and, additionally, roll ‘yes’ on the boolean dice. This rule helps to balance the gameplay, by preventing the player from getting too many pieces onto the board too quickly. Once the Swallow piece entered the board, the player can roll their boolean dice, which will convert the number obtained from the number dice. If the boolean dice indicates ‘yes’, numbers 1, 2, 3 and 4 on the numbered dice are converted to 5, 6, 7, and 10, respectively.

Once at least one piece is launched, the player moves their piece by the number of squares indicated by the numbered dice. The paths of each player takes from the start to the finish line are initially separate, just to overlap, and then split again at the end. In the overlapping fragment of the path the player’s piece may be kicked-out by the opponents pieces. This happens if a piece lands in a square occupied by the opponent’s piece, that piece will be knocked off the board. The knocked-off piece can be re-launched in accordance with the rules described above. The game ends when one of the players manages to get all their pieces to the end of the game. [1].

3 IMPLEMENTATION

We implemented the “Royal Game of Ur” for Oculus Quest Virtual Reality Headset. We used Unity 3D game engine and C# programming language.

3.1 Game Environment

The games starts with a player finding themselves locked in the tomb of the Sargon of Akkad, a ruler of the Akkadian empire who is believed to be an avid player of the game. The only way to exit

the tomb, and therefore to save themselves, is to win a Royal Game of Ur game against an the stone sculpture of the Sargon of Akkad. The environment of the game closely resembles the interior of the tombs in the city of Ur, with reliefs and sculptures modelled after the original Akkadian reliefs.

3.2 Implementation of the Game's AI

To equip Sargon of Akkad with a skill of playing the game, we implement the rules of the game using MinMax algorithm. Since the resulting search tree would be too deep for a computational capabilities of most of the standalone VR or AR headsets, we used alpha-beta pruning technique. To further increase players chances of winning, we introduce random, non-optimal choices for the AI.

3.3 Game Interaction

Instead of using the Quest Touch controllers, for player's interaction with the pieces and dices we decided to experiment with the new Quest hand-tracking. The player can interact with the objects in game by performing hand gestures such as 'gripping' to roll the dice, and 'pinching' to pick up and move the pieces.

As an alternative form of input, we experiment with NextMind brain-computer interface to allow for derogating experience, where the player picks up and moves the pieces and rolls the dices using their brainwaves as input. In our experiments with NextMind devices, as long as the game elements are large enough (about 2x2 cm in the VR view), target acquisition using EEG analysis of the visual cortex is reliable, although the processing takes about 2 seconds. Since the "Royal Game of Ur" is a board game, the latency does not affect the player's chances of winning. The effect of latency on the game ergonomics remains our future work.

3.4 Haptic Feedback

To provide for increased immersivness, we experimented with passive haptic feedback by allowing the users to play the VR game on a real board. The tracking of the board and the pieces was implemented using Aruco markers. The original size of physical pieces and dices are too small to be reliably tracked with Aruco code. Also, tracking may be lost due to the Aruco codes being occluded by the hand of the player. In the future, we plan to replace the Aruco code tracking with magnetic tracking.

4 EVALUATION

The game was evaluated during the Student Game Festival, where twenty players had an opportunity to play the game for few minutes. Their feedback was generally positive, with some players complaining about the precision of the hand tracking. At the time

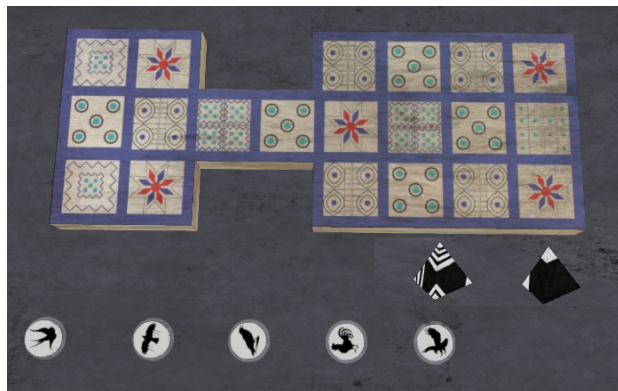


Figure 2: The board, the pieces, and the dices

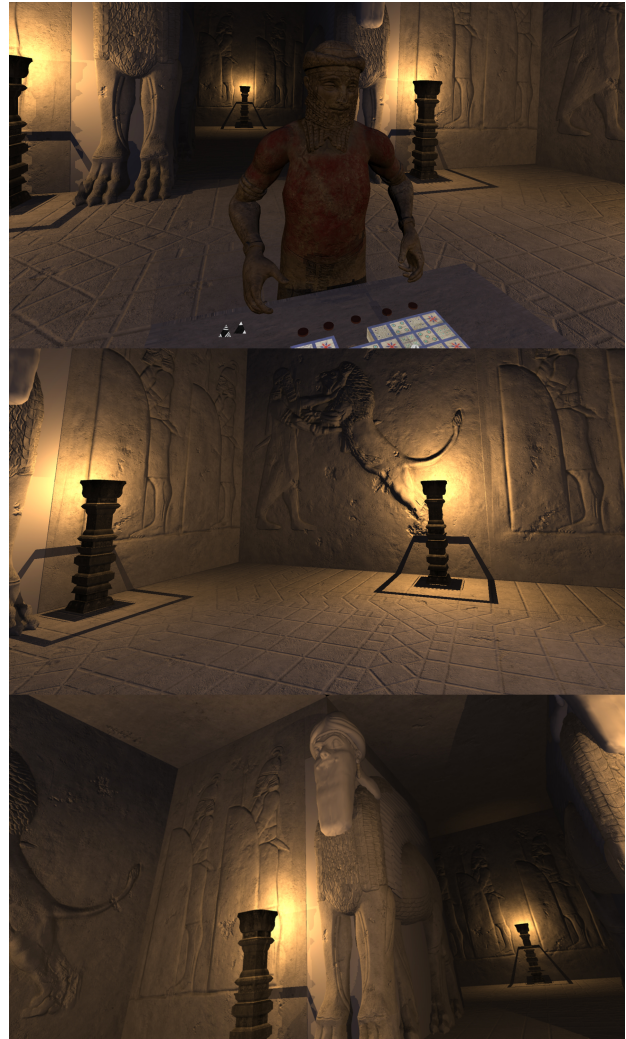


Figure 3: Game Environment – realistic recreation of the Tombs of Ur and the statue of Sargon the Great

of evaluation, we were still using Leap Motion for freehand interaction. Users' feedback motivated us to update the system from the originally using Leap Motion to the new, more reliable Oculus Quest hand tracking. However this new implementation was not yet evaluated outside of our research group.

5 ACKNOWLEDGMENTS

We would like to thank Pawel Synowiec and Grzegorz Dabrowski for the artwork used and Morgan Johnson, Daniel Kharlamov, and Ryan Blakeman for their help in developing the game.

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